EduGenie: -Project Development Phases

1. Brainstorming & Ideation Phase

The brainstorming and ideation phase is the initial stage of the EduGenie project. The main objective was to identify a common problem faced by students during their learning process and develop an AI-based solution.

Students often find it difficult to understand complex topics, summarize lengthy study materials, prepare quizzes, and identify the correct learning resources. To address these problems, the idea of EduGenie – Google Gemini Powered Learning Assistant was proposed.

EduGenie is designed as a lightweight educational assistant that uses Generative AI to provide interactive learning support.

1. **Main Ideas Identified**

- Provide instant answers to students' questions.
- Explain difficult concepts in simple language.
- Generate quizzes automatically.
- Summarize lengthy educational content.
- Provide personalized learning paths.
- Help students learn from beginner to advanced levels.
- Provide a simple and user-friendly interface.

2. **Proposed Solution**

The proposed solution was to develop a web-based AI learning assistant using:

- Python for backend development.
- FastAPI for REST API development.
- HTML and CSS for the frontend.
- Google Gemini API for generative AI features.
- LaMini-Flan-T5 for lightweight concept explanation.

The idea was selected because it combines artificial intelligence with education and can be used by students with minimal technical knowledge.